

CA Note

1 Basic Knowledge

Group $(G, *)$: **G1**: Closure **G2**: Associativity **G3**: Identity element **G4**: Inverse element

Ring $(R, +, \cdot)$: **R1**: $(R, +)$ is an abelian group **R2**: $(R, \cdot)$ is monoid. (no inverse) **R3**: Distributive law (left and right)

Ideal (I) : **I1**: $\neq \emptyset$ **I2**: $\forall a, b \in I, a - b \in I$ **I3**: $\forall r \in R, i \in I, ri, ir \in I$ **Lemma**: $I = R \Leftrightarrow 1 \in I \Leftrightarrow I \cap R^\times \neq \emptyset$

Subgroup (H) : **H1**: $e \in H$ **H2**: $\forall a, b \in H, ab \in H$ **H3**: $\forall a \in H, a^{-1} \in H$ **Subring** (S) : **S1**: $1 \in S$ **S2**: $\forall a, b \in S, a - b \in S$ **S3**: $\forall a, b \in S, ab \in S$

Integral Domain (ID): **ID1**: commutative ring **ID2**: no zero divisors **Lemma**: Finite ID $\Rightarrow$ field ID $\Rightarrow$ Cancellation Law

Factor Ring (R/I) : $a + I = b + I \Leftrightarrow a - b \in I$ $(R/I, \oplus, \odot)$ is a ring. ($1_{R/I} = 1_R + I, 0_{R/I} = 0_R + I, I$ ideal)

Ring Homomorphism: $\varphi : R \rightarrow S$ s.t. **RH1**: $\varphi(a + b) = \varphi(a) + \varphi(b)$ **RH2**: $\varphi(ab) = \varphi(a)\varphi(b)$ **RH3**: $\varphi(1_R) = 1_S$

· **1**: $\text{Ker}(\varphi)$ is an ideal. **2**: $\text{Im}(\varphi)$ is a subring. **3**: φ is 1-1 $\Leftrightarrow \text{Ker}(\varphi) = \{0_R\}$. **FIT**: $R/\text{Ker}(\varphi) \cong \text{Im}(\varphi)$ (First Isomorphism Theorem)

· **Canonical Homomorphism**: $\text{can} : R \rightarrow R/I$ s.t. $\text{can}(r) = r + I$ is a onto ring Homomorphism. $\text{Ker}(\text{can}) = I$

· **Lemma**: If $\varphi : R \rightarrow S$ ring Homomorphism s.t. $I \subseteq \text{Ker}(\varphi)$, then $\exists! \tilde{\varphi} : R/I \rightarrow S$ s.t. $\varphi = \tilde{\varphi} \circ \text{can}$ (i.e. $\tilde{\varphi}(r + I) = \varphi(r)$)

Lagrange's Theorem: G finite group, $H \leq G$ subgroup, then $|H| \mid |G|$

2 Types of Ideals